“How Brains Work”
(or: some metaphors from neuroscience that are important for psycholinguistics...)

LING/PSYC 406 Session 4
(Covering Menn, Chapter 2)

Association Norms:

We've read a little about these.

For your first taste of what it's like to be a participant in one of these studies (or maybe not if you've done research for extra credit before) we'll do some "norming" in class.

I'll put a word on the screen for a few seconds, and you write down the first few words that come to mind.

LION

TRIPE

FREE

STRIPES

PEAR

BEAR

TREE

TIGER

Association Norms:

We're not going to tabulate norms for all of the items. Let's do three that will be critical for later:

LION

TIGER

STRIPES

(I was making a point about the arbitrariness of form-meaning relationships by including pairs like TRIPE/STRIPES, PEAR/BEAR)

Association Norms:

OK. That was fun.

Think about your experience coming up with associates.
Did you have a particular strategy?

Do you think the list structure could have influenced the answers you provided? What if I'd only asked about wild animals? Or if instead only listed household appliances?

Look at the "norms" we produced:

How is "associated" different from "similar?"

“How Brains Work”

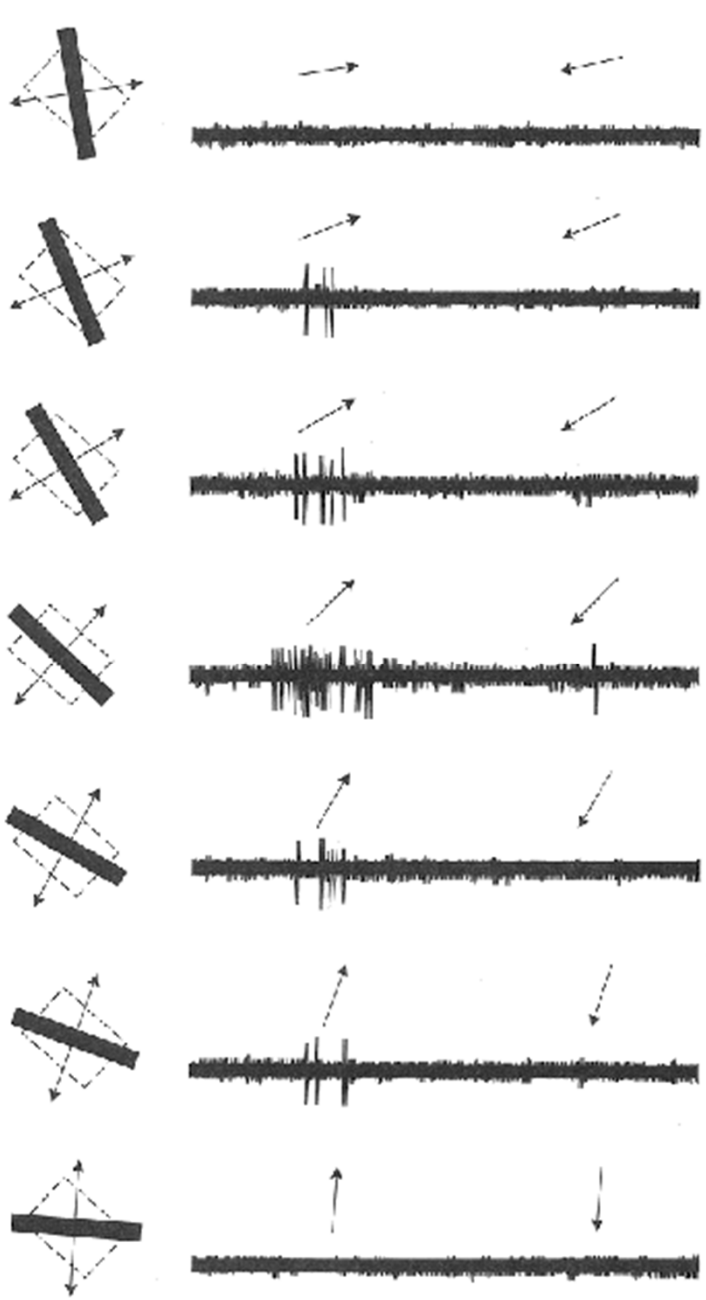
What an amazing thing to call a chapter!

A better title might have been: *Some basic metaphors from neuroscience that are useful in psycholinguistics.*

We’ll go through these one by one, because they really are the basis for understanding most theory (and experimental design, for that matter) in the field:

- Activation
- Spreading of Activation (over weighted connections)
- Distributed Representations
- Associative (e.g., Hebbian) Learning
- Top-down and Bottom-up Processing

Activation



Hubel & Weisel (1968)

Contemporary with the Cognitive Revolution, (which we'll talk more about soon) much important work was done on relating neural activity to sensory input.

Here we see simple cells in primary visual cortex (V1) responding to lines of particular orientations in a particular **receptive field**.

Electrochemical **activity** in the cell tells us about its **representational** and **computational** properties.

This cell activates most strongly to upward movement of a line at an angle of about 315°, so we can talk about this cell as **representing** that combination of features.

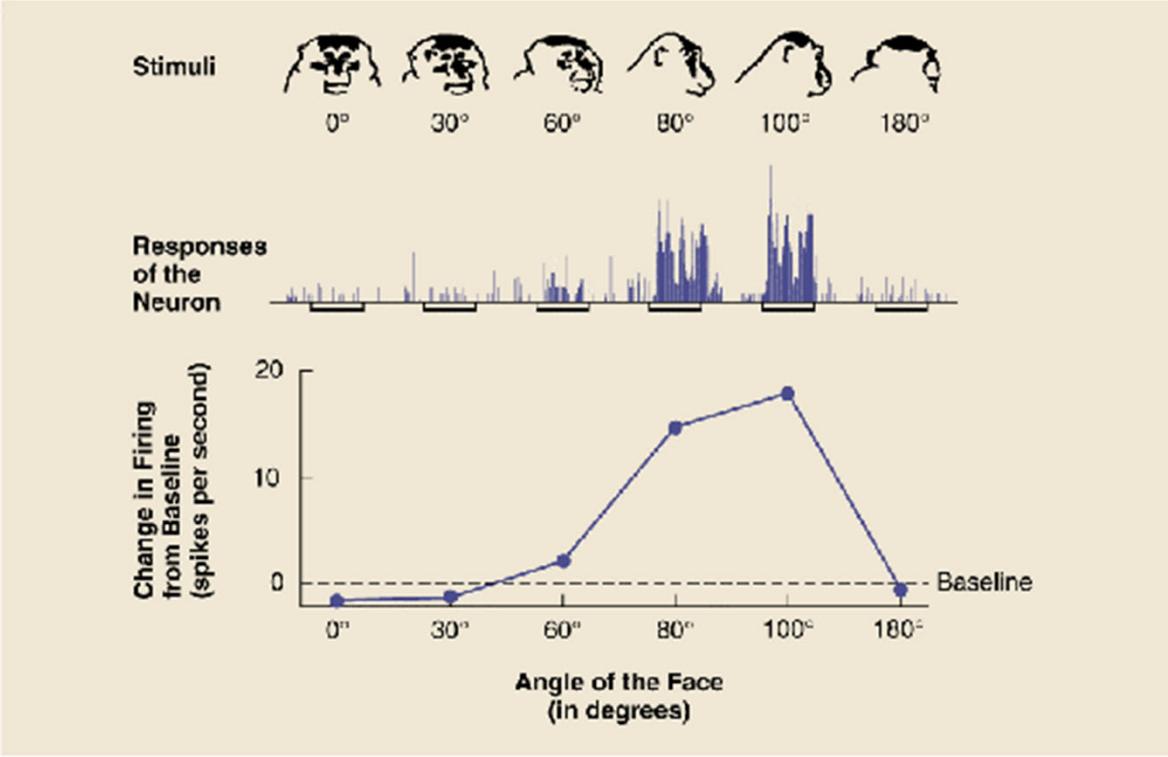
Activation

Here's a much cell from "further up" in the "visual processing stream."

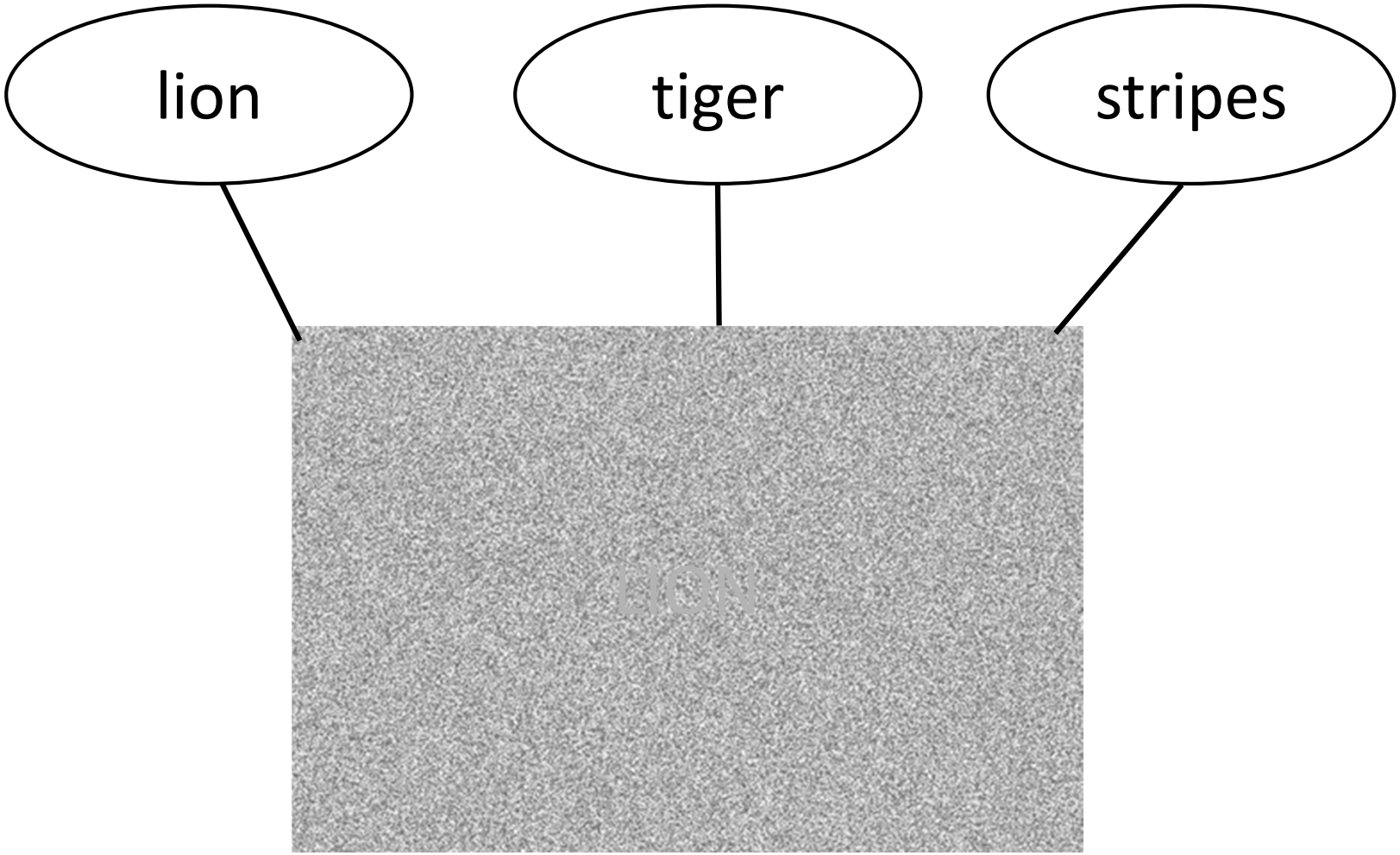
It seems natural to think of this cell as "representing" something more "complex." (Like the profile of another monkey, vs. line segments)

"Activation" in psycholinguistics extends this (metaphorically) to perceptual and conceptual representations.

► Firing Rate of a Monkey Inferotemporal Cortex Neuron

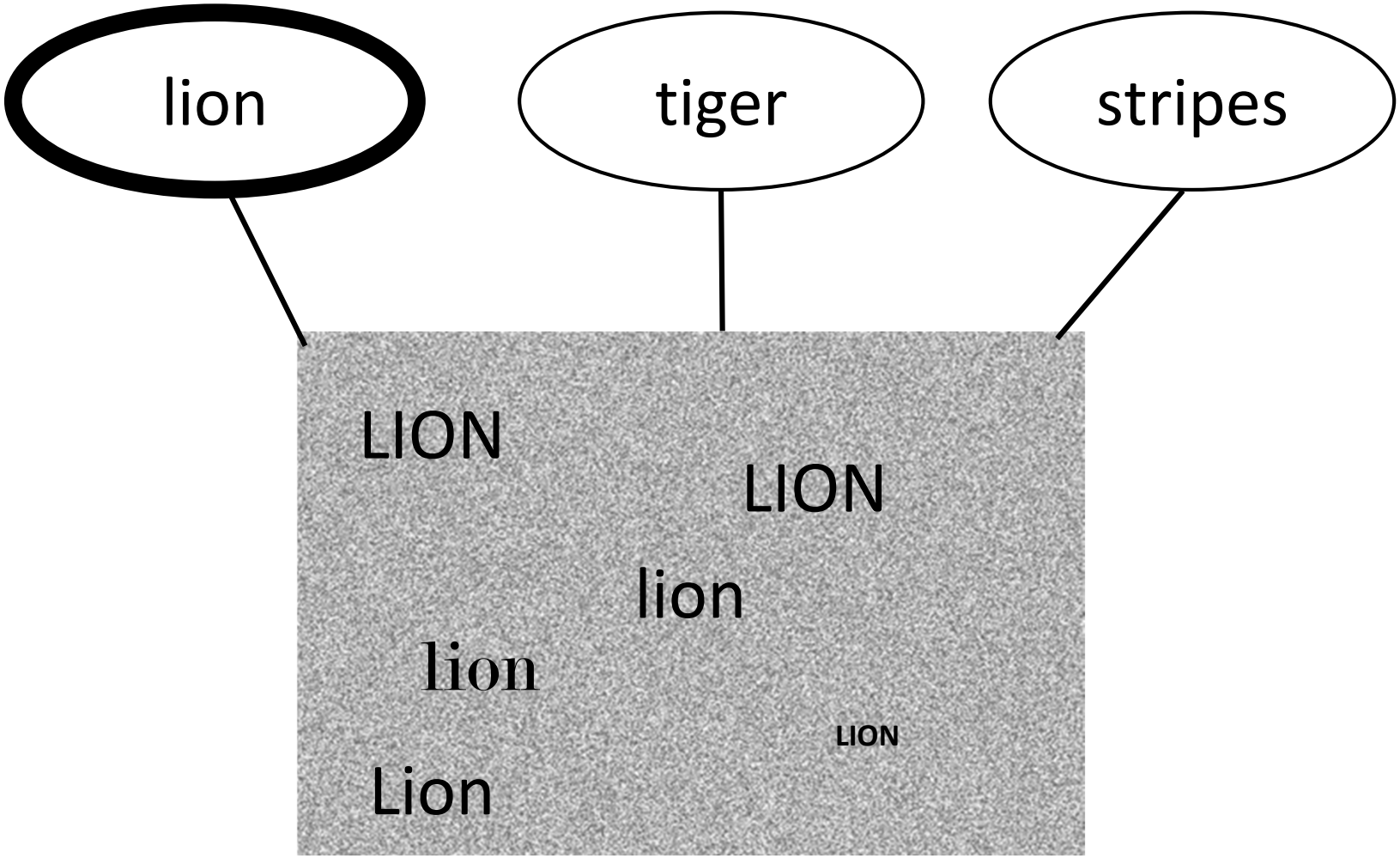


Source: Adapted from Gross et al., *Pattern Recognition Mechanisms*, p. 179-201. Berlin: Springer-Verlag 1985.



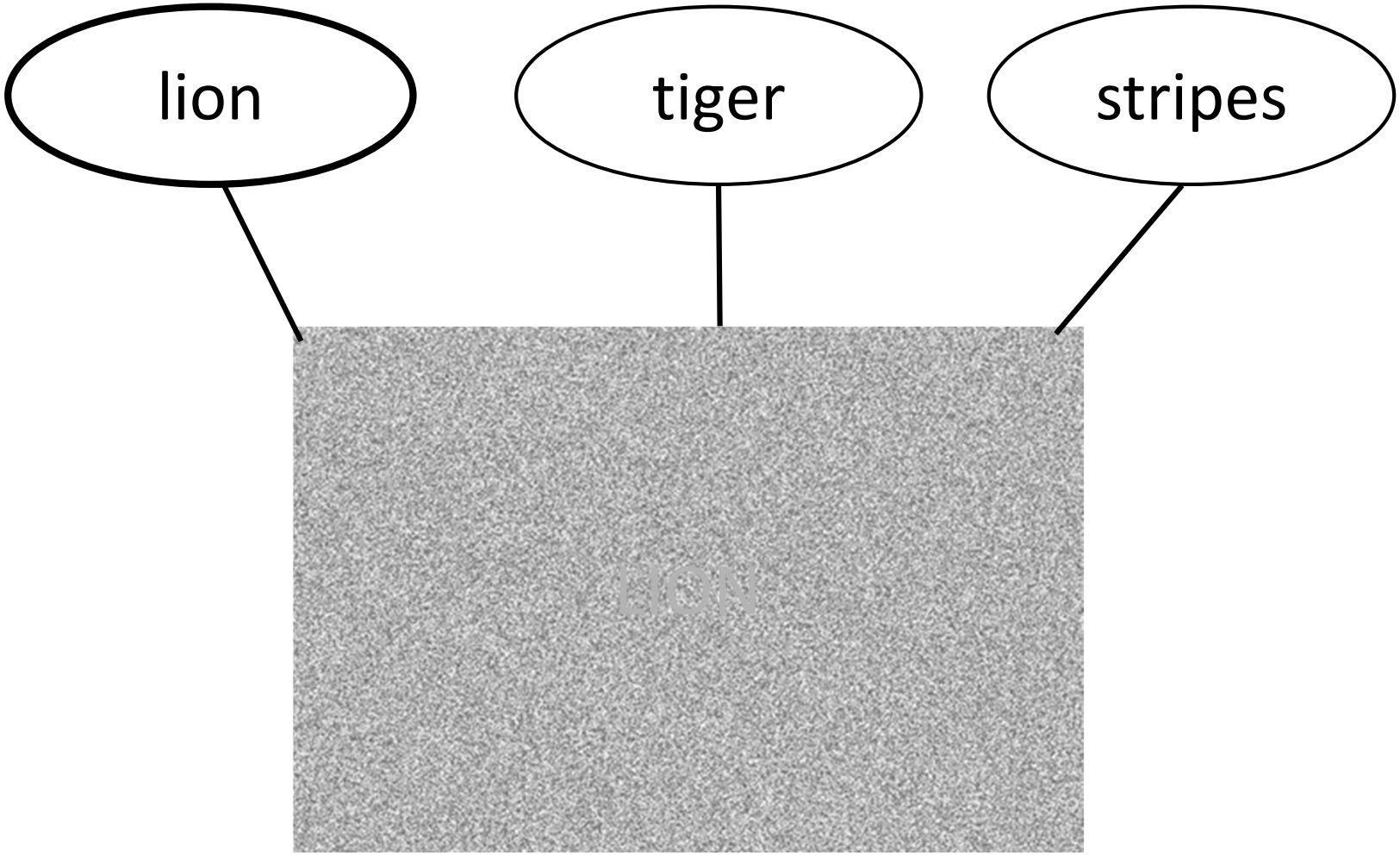
Let’s imagine an incredibly simple set of **lexical representations**.

These would be **nodes** in a **network** that can be **activated** by a specific word.



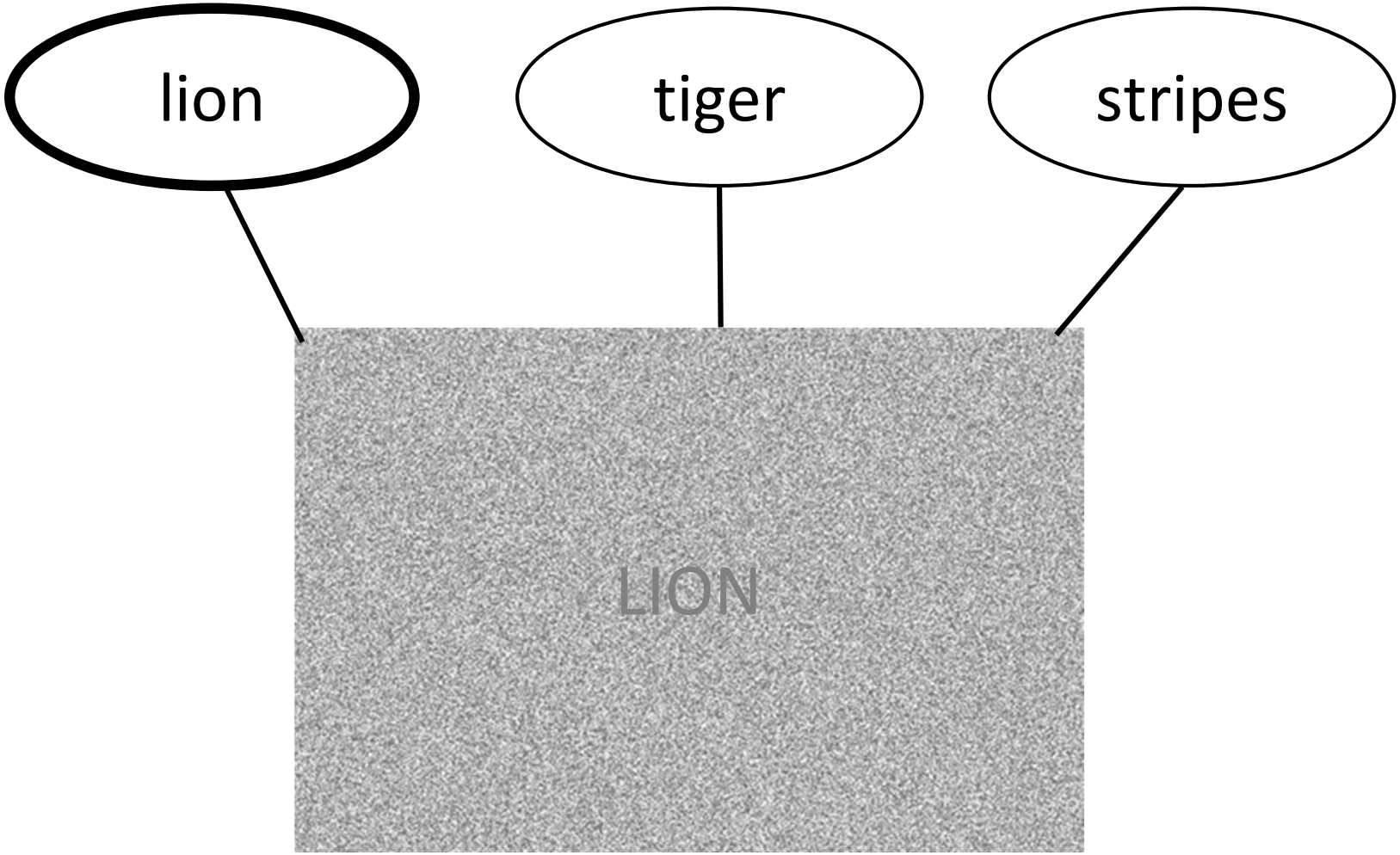
Eventually we'll have to consider how complicated the "receptive fields" for these nodes would have to be to deal with the **lack of invariance** problem (again, remember this with phonemes).

For now, just imagine there are nodes that can be activated by words.

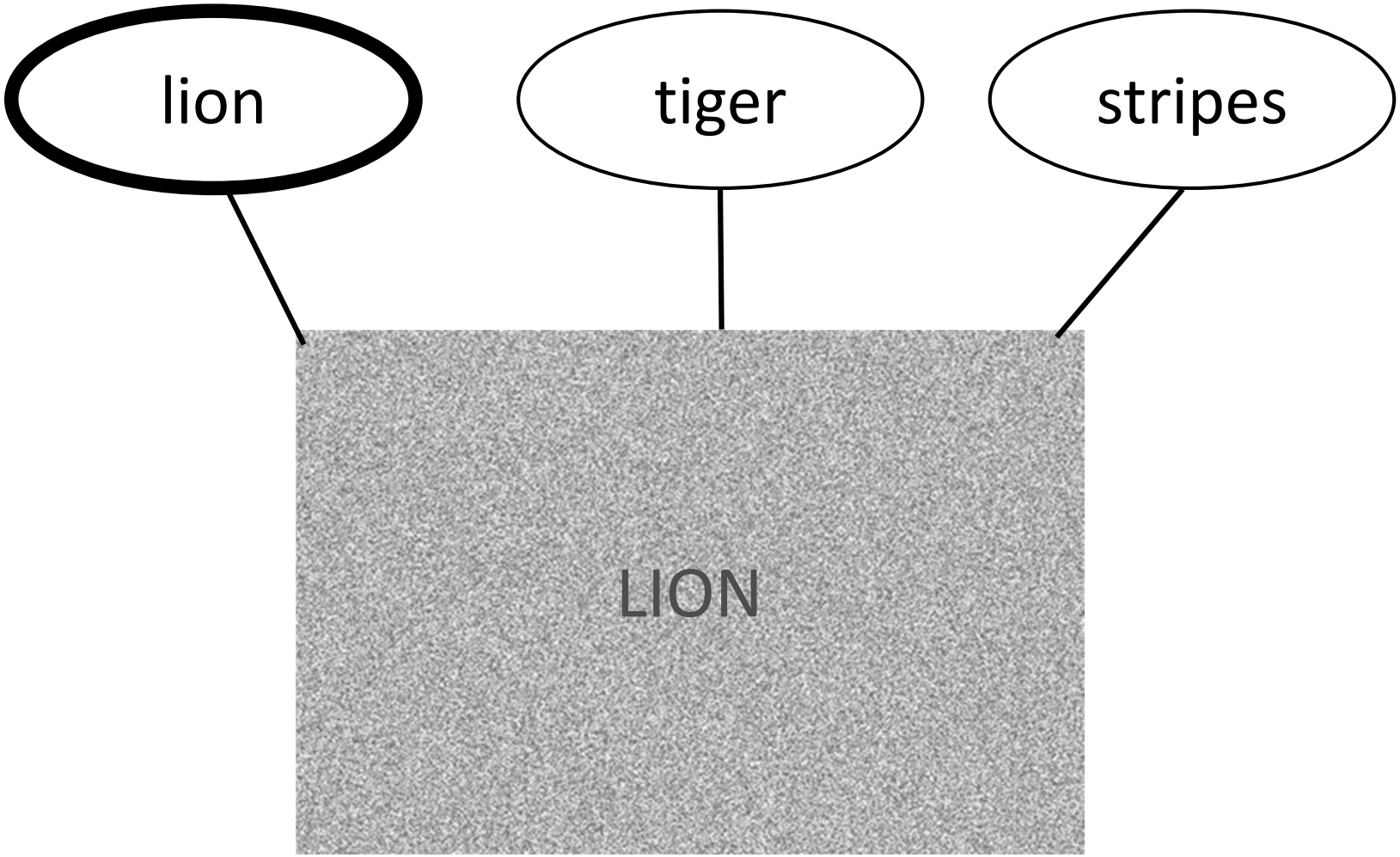


Activation can be partial.

Here, the word LION contrasts poorly with the noise mask, so we might imagine only weak activation as a result.



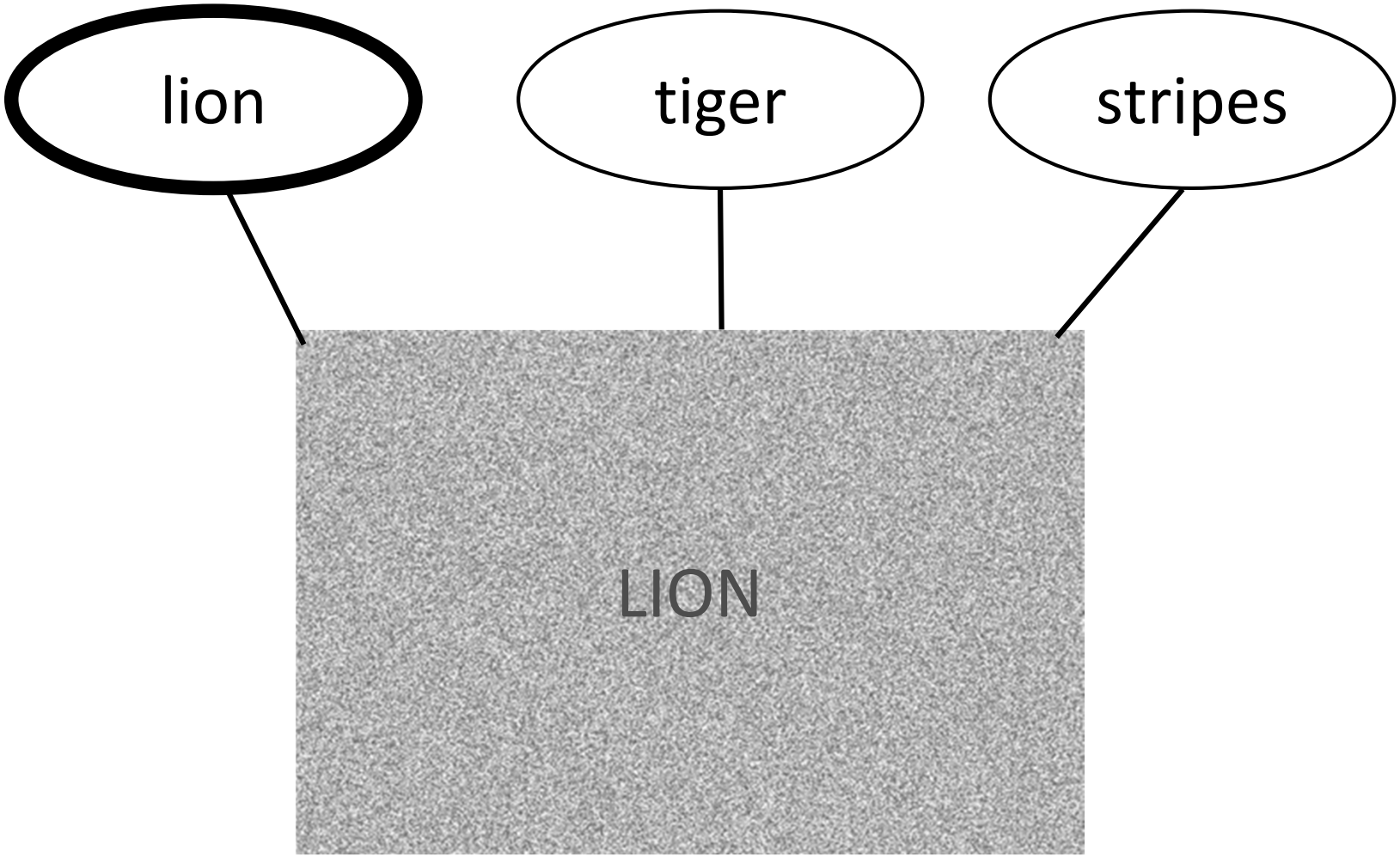
As the stimulus becomes clearer, activation increases.



At some point, the signal-to-noise ratio (SNR) is very favorable.

Activity is strong enough that we can say this simple network has “identified” the word “lion.”

In fact this is an illustration of the “perceptual identification task,” in which participants are asked to push a button as soon as they can identify the item.



What factors might influence how quickly you can identify a word under conditions like this? Consider:

features of the stimulus itself
features of the experimental environment

What factors might influence how quickly you can identify a word under conditions like this? Consider:

features of the stimulus itself

- how frequently a word is encountered
- *how similar it is to other words*

features of the experimental environment

- whether you just saw that same word (repetition **priming**)
- *whether you just saw a similar or related word...*

Activation dynamics

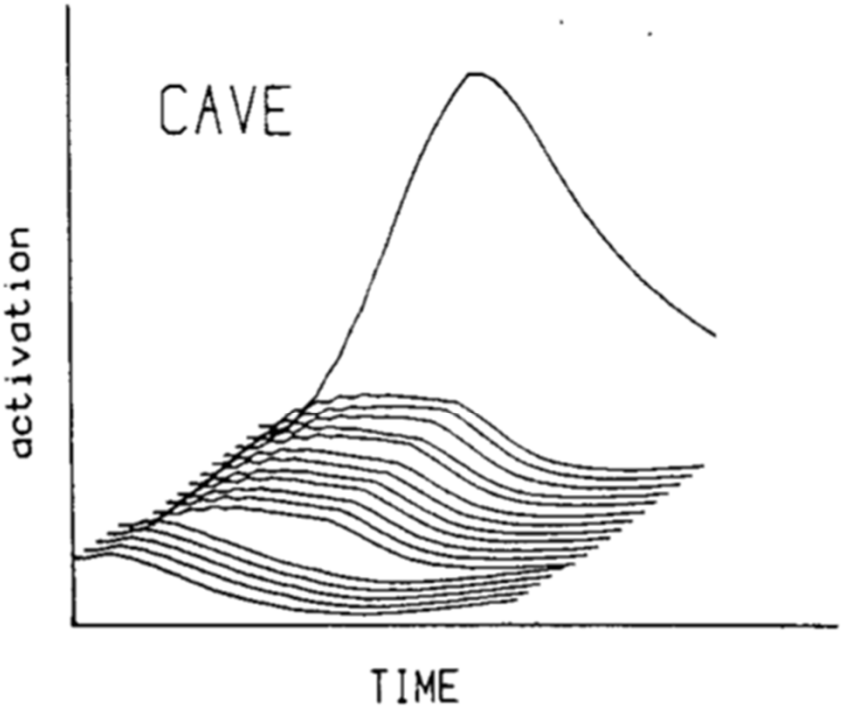


Figure 13. Activity at the word level upon presentation of *CAVE*, with weak letter-to-word inhibition.

Different computational models (like Rumelhart & McClelland's 1986 Interactive Activation model shown here) simulate frequency effects in different ways

Most models of this kind assume that activation unfolds over time.

One way to explain **frequency effects** is to assume that **activation dynamics** are tuned up by experience.

Words that are encountered more often get tuned up to be more sensitive, and are easier to activate.

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Activation dynamics

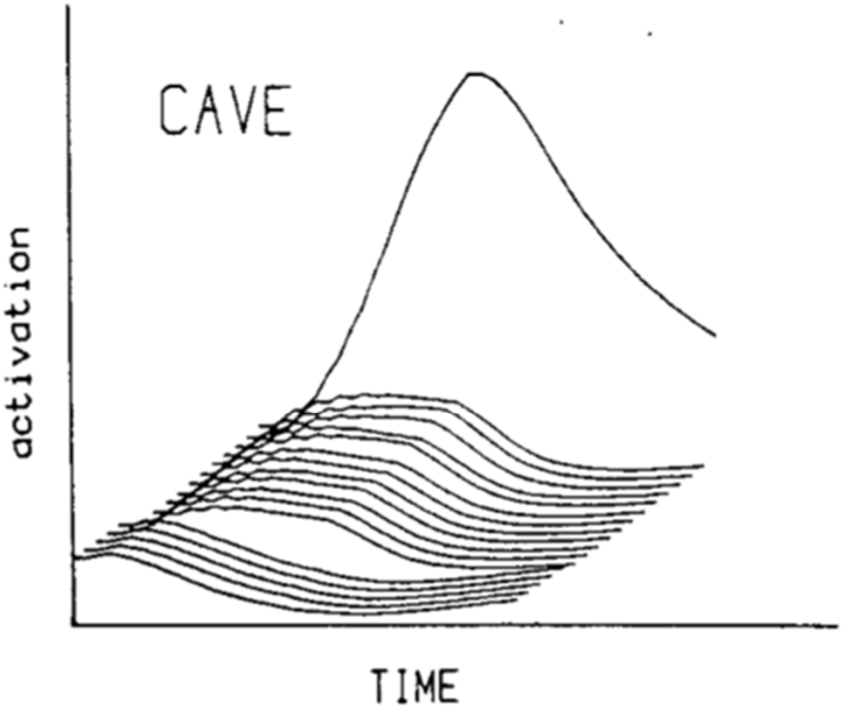


Figure 13. Activity at the word level upon presentation of *CAVE*, with weak letter-to-word inhibition.

We can also explain why seeing a word again, very shortly after just having seen it leads to fasted identification in these terms.

Activation does not rise and fall instantaneously.

We might assume (as shown) that it falls off more slowly than it rises.

If the word is encountered again while activation is still high from the last exposure, the “baseline” for the activation is already higher than usual, so it can be activated more easily.

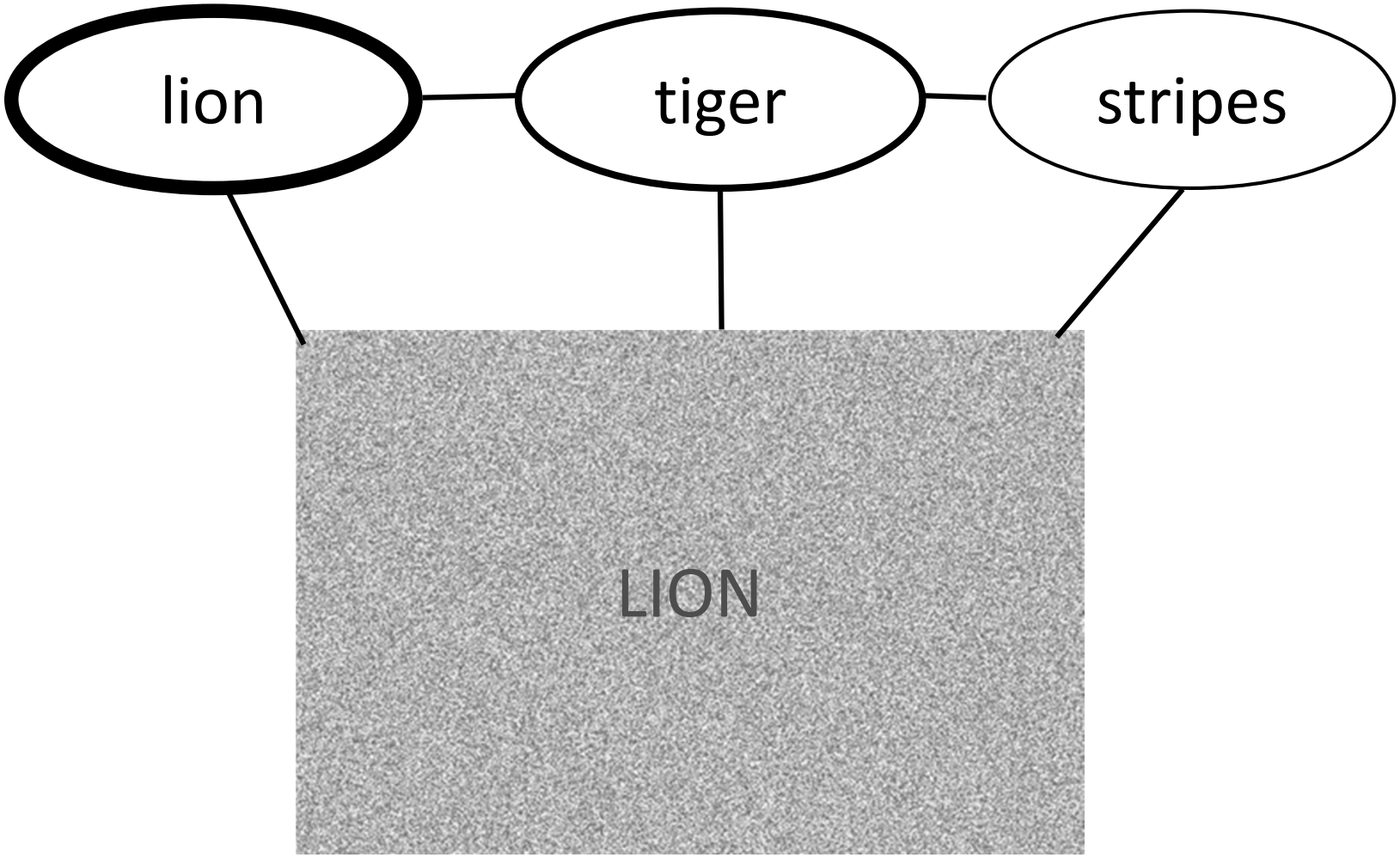
This is *repetition priming,* a basic effect in psychology.

So far I've left out two factors that can influence a word's activity:

how similar it is to other words (neighborhood effects)

whether you just saw a similar or related word
(associative/semantic/phonological priming)

...because they require us to add something to our little network.



Understanding many phenomena in language requires assuming that nodes in a network are **connected.**

(Which, duh, it's a *network.)*

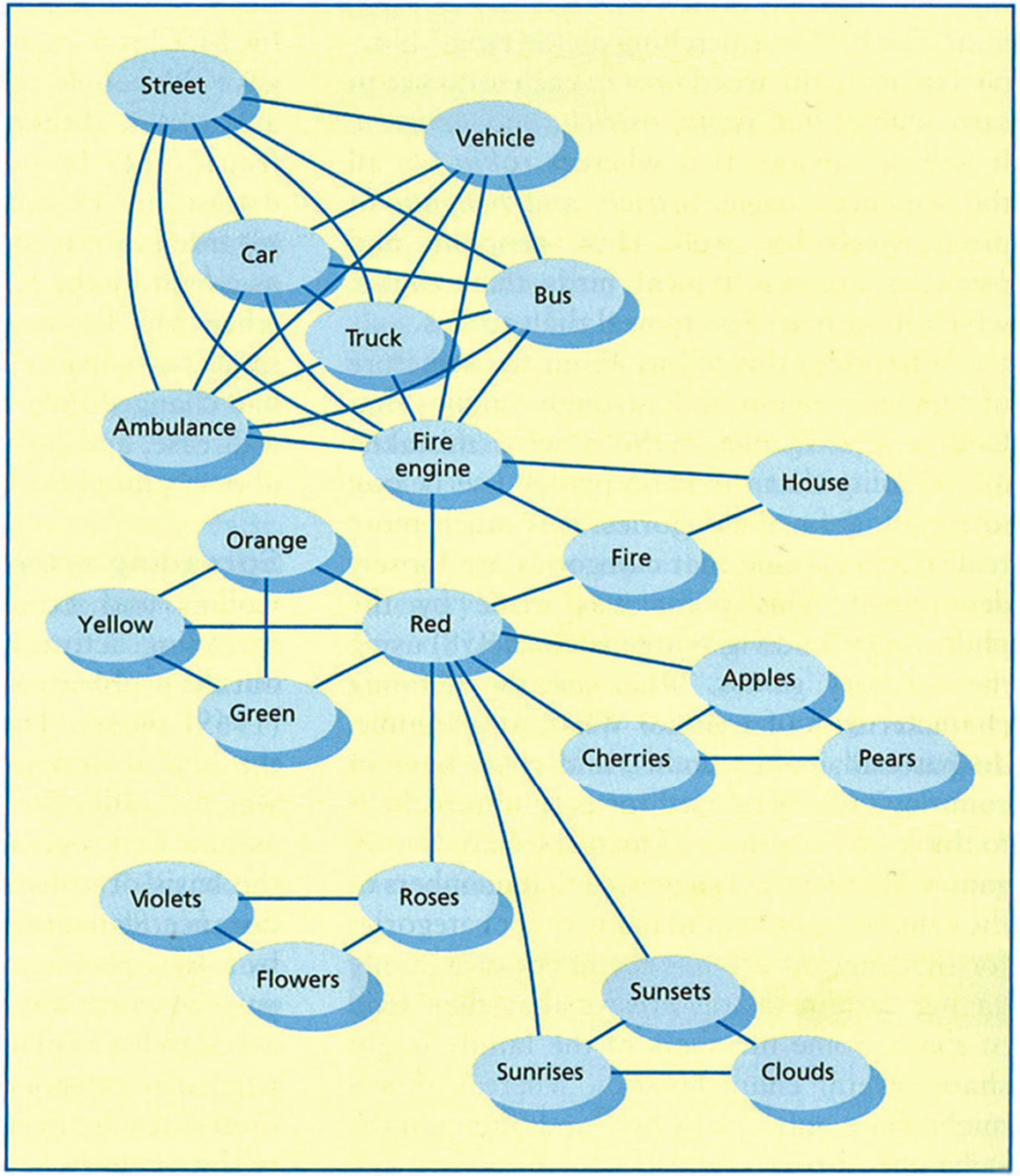
Spreading Activation

Remember our norms?

One way to think about how those associations are represented is to imagine a network in which the connection strengths between nodes are determined by their association).

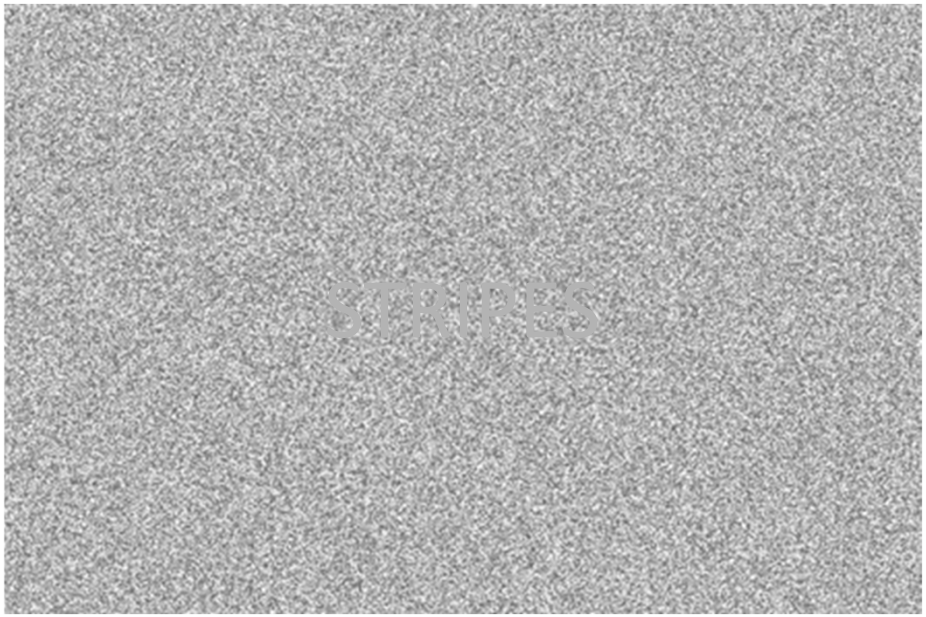
So when you activate “lion,” some of that activation travels along a **weighted connection** to “tiger.”

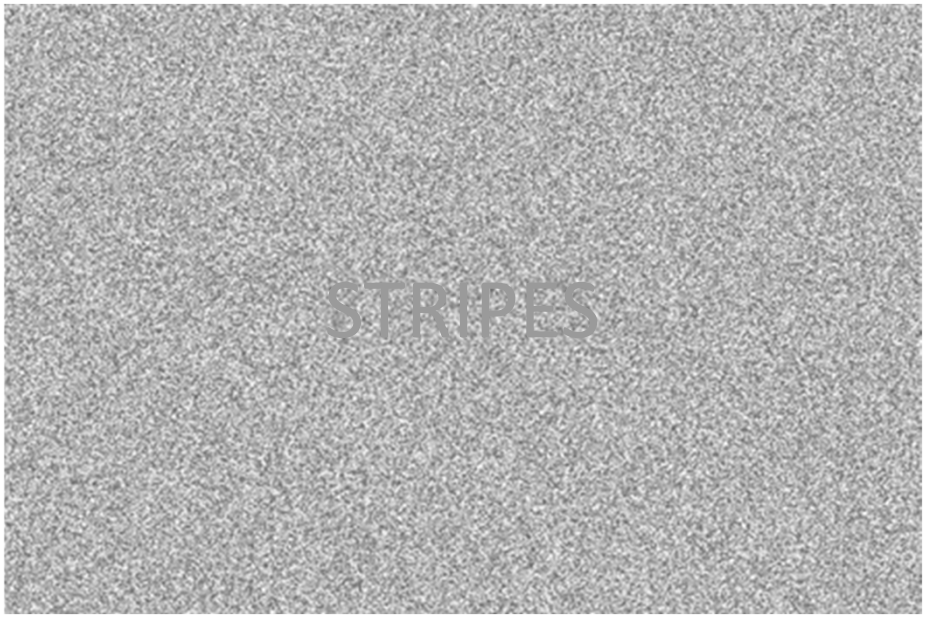
This is a spreading activation account of semantic priming.

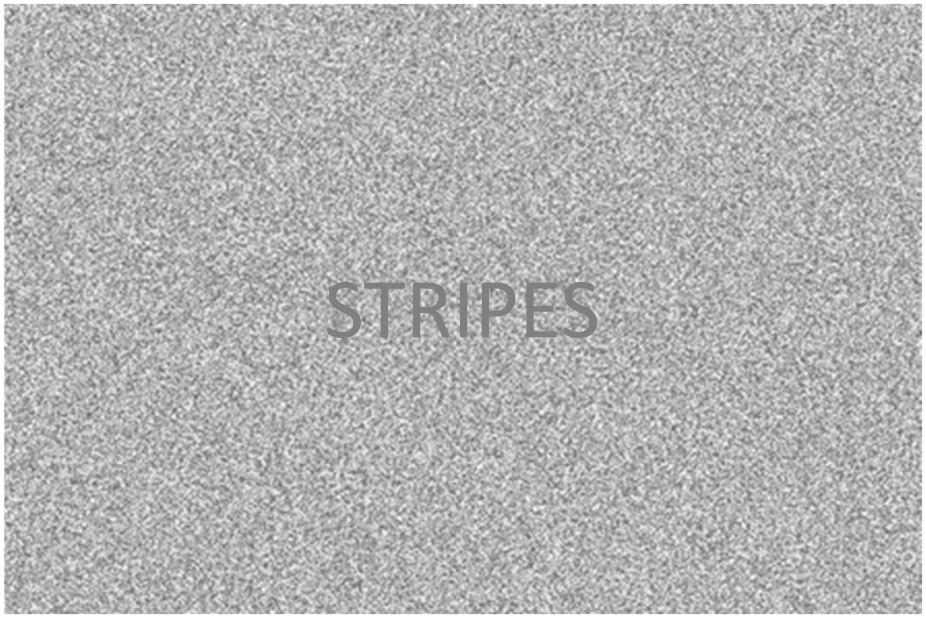


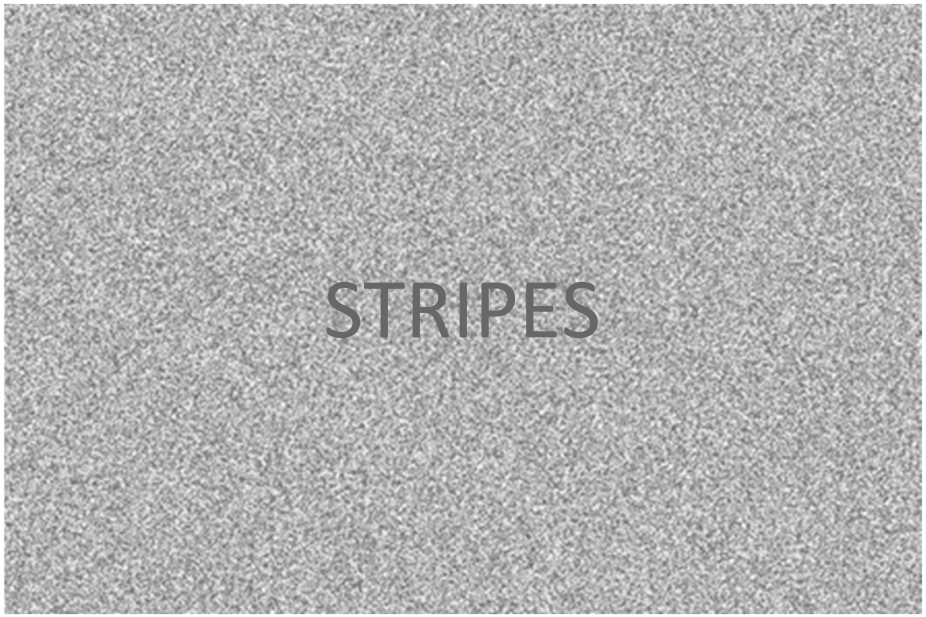
Graphic from this lecture by Alcom and Thomas:
http://www.inf.ed.ac.uk/teaching/courses/inf1-cg/lectures/12/

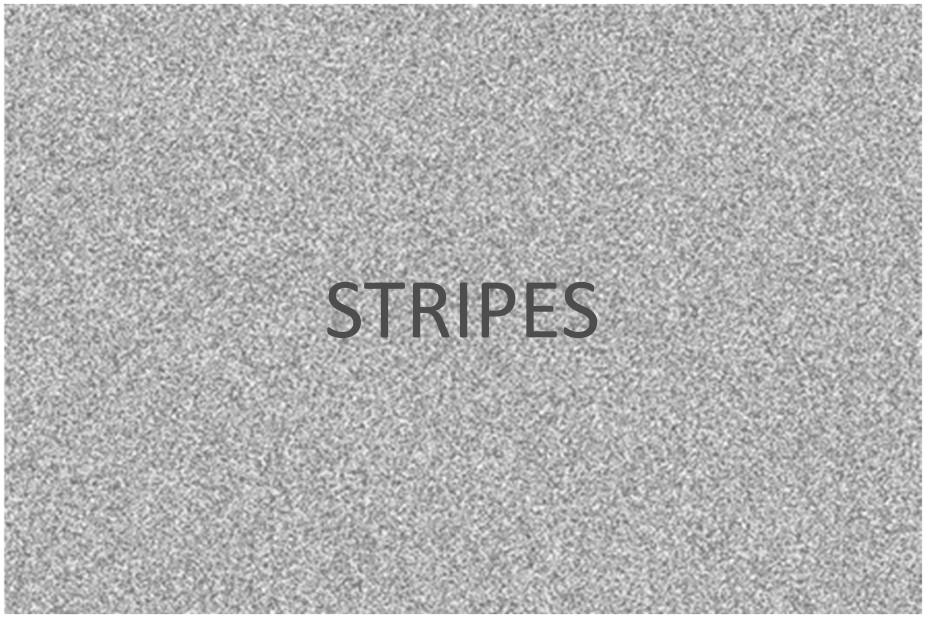
A neat experiment...

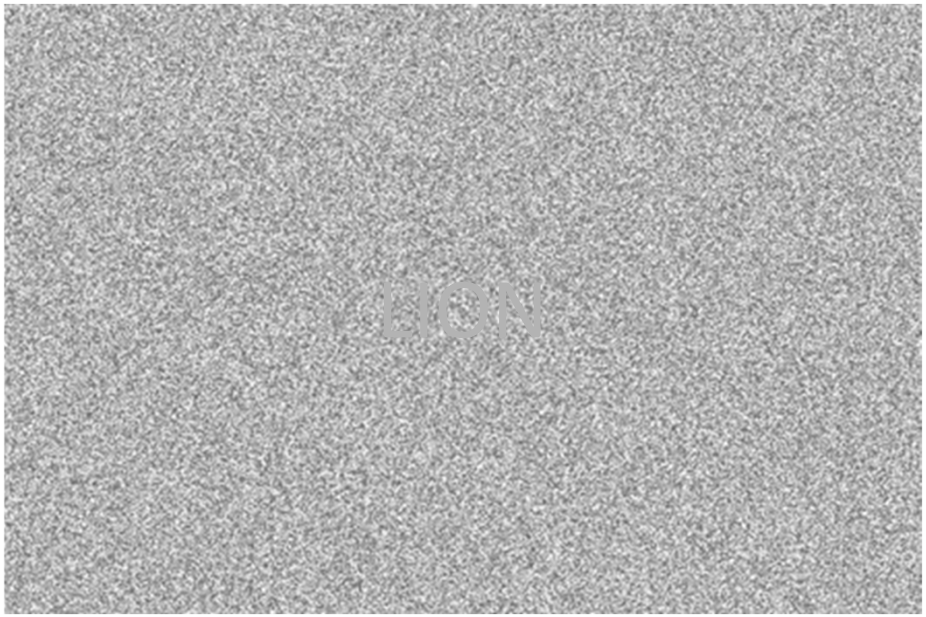

STRIPES

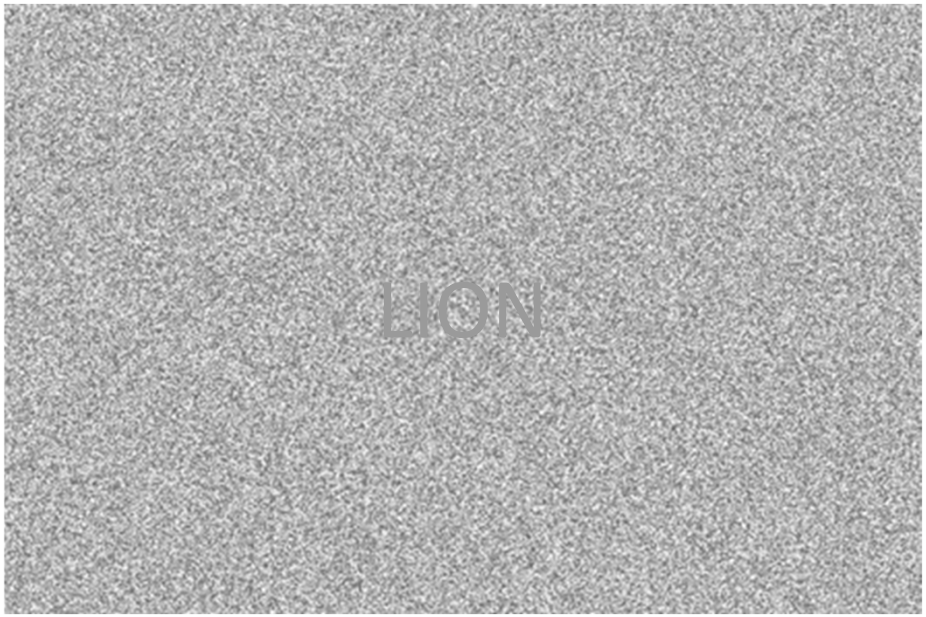

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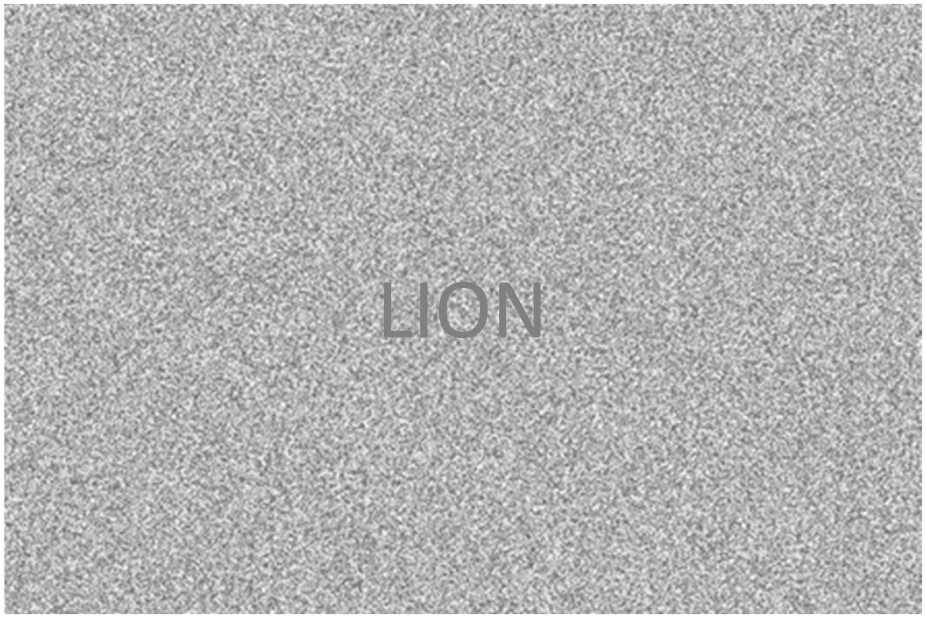

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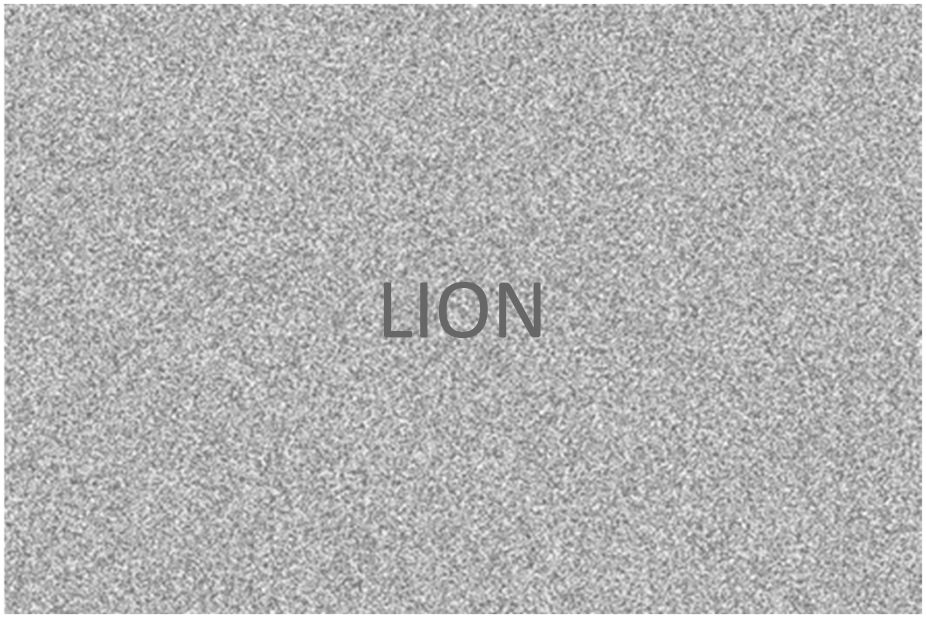

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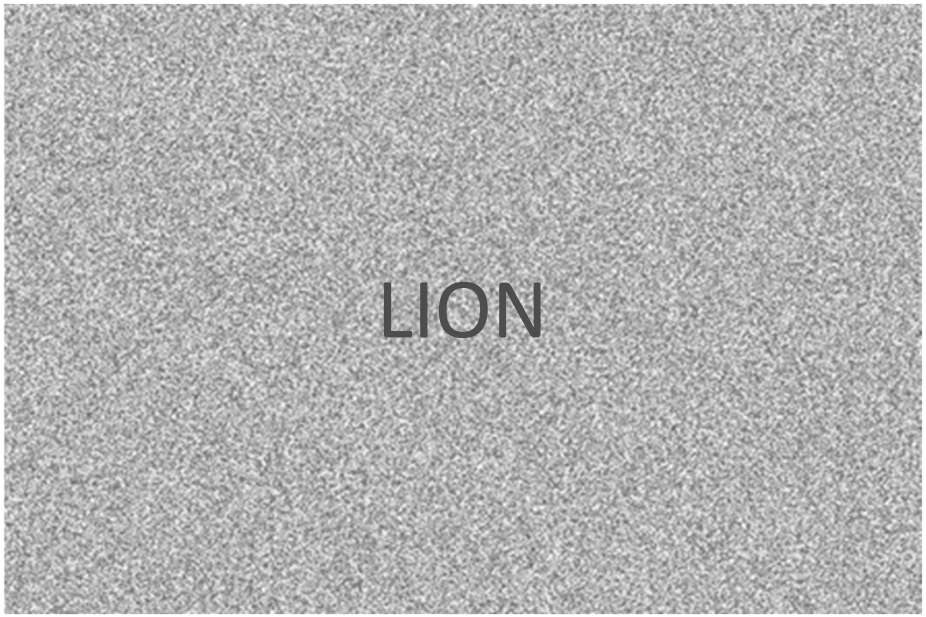

STRIPES

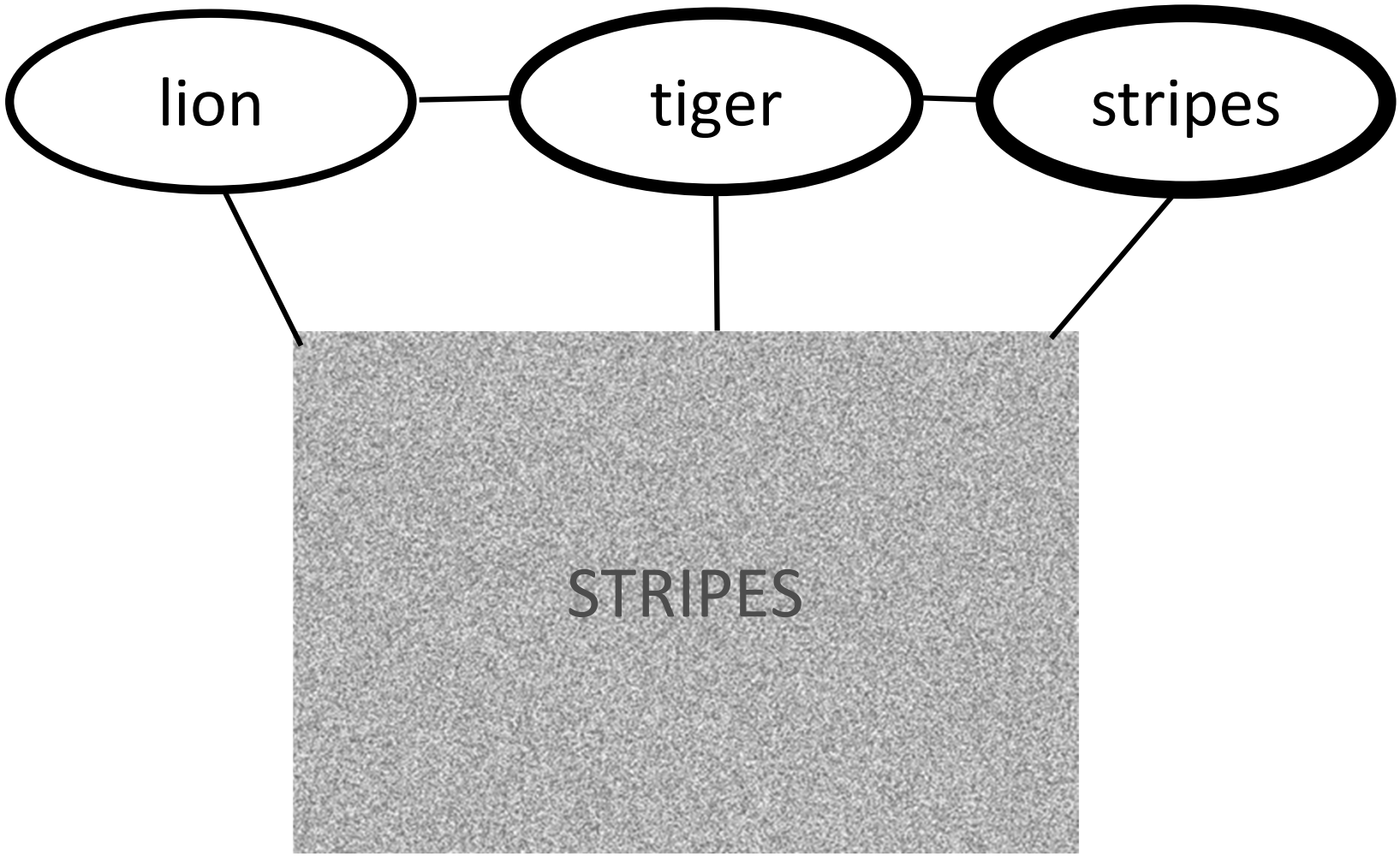



LION


LION


LION


LION



The idea is that activity flows throughout the network, so activating “stripes” can prime lion, even though these are not *directly* associated.

This is called *mediated priming*. It provides compelling evidence for the notion of *spreading activation*.

This is very representative of a lot of psycholinguistics:

careful study design

small effects

Counterintuitive: why would “stripes” prime “lion?”

...now we will briefly consider three more “big picture” issues

Bottom-up and top-down processes.

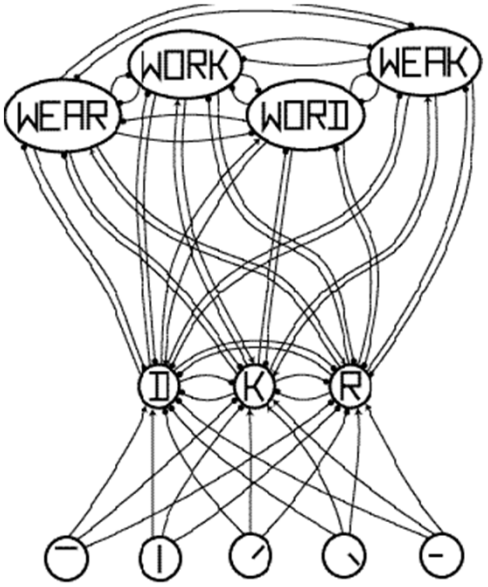


Figure 1. Interactive Activation Network Model (after McClelland and Rumelhart, 1981).



Figure 5. A hypothetical set of features that might be extracted on a trial in an experiment on word perception.

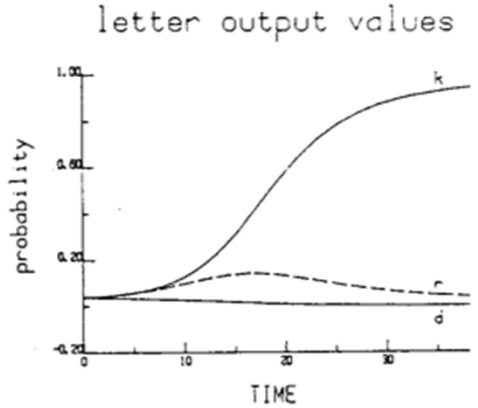


Figure 7. Output values for the letters *r*, *k*, and *d* after presentation of the display shown in Figure 5.

Top-down processing, whereby activation for WORK makes the last letter here look more like an R than a K, is one way of modeling “pattern completion,” which Menn discusses at length.

Learning:

How does all of this stuff get into the brain?

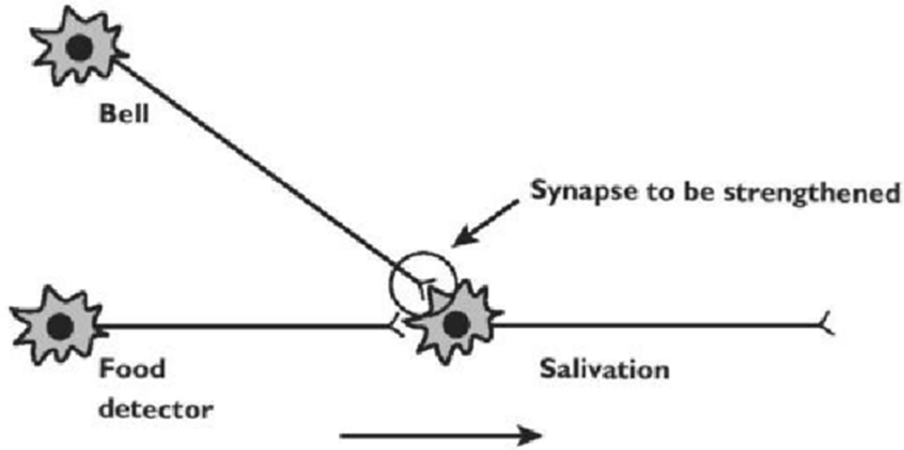
An extremely simple, but powerful idea...

"Neurons that fire together, wire together."

Of course it is more complicated than this, but an important property of the neural learning mechanisms that are relatively well understood is that they are basically *associative*.

Historically, these mechanisms were considered insufficient to learn anything important about language.

More recent arguments are about what role they might play, and how they might be constrained by innate knowledge. (Especially given that LLMs can learn all kinds of structural features of language without much more than this.)



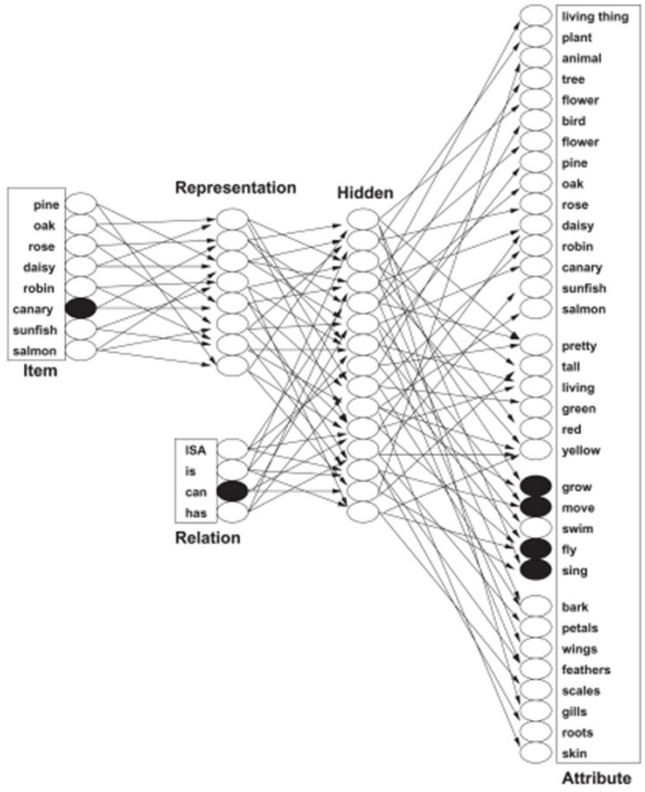
Graphic from this nice primer on neuroscience of motivation (by a philosopher, I think):
http://lesswrong.com/lw/71x/a_crash_course_in_the_neuroscience_of_human/

Distributed Representations:
How associative learning mechanisms can learn relatively sophisticated patterns.

Remember we talked about how important *generativity* is for language?

In order for an associative model to *generalize*, it has to be able to represent similarity.

So, what you know about robins and canaries can generalize to new birds as you learn about them.



So, we've learned some neurally-inspired metaphors for talking about psycholinguistics:

- **activation**
- **spreading activation**
- **top-down** and **bottom-up** processing
- **associative** (e.g., Hebbian) **learning**
- **distributed representations**

We also began to look a little bit at how people have used *priming* experiments to reason about what people's lexical networks might look like.

For Thursday

Two readings (on Brightspace):

- an intellectual biography of Paul Broca (by Stanley Finger)

- a review of the state-of-the art of neuroimaging circa 1988 (Posner et al.)

Extra Stuff:

If you are coming to this course from a linguistics background, or haven't thought about different theories of mem

Note that many of the assertions there are oversimplifications, but the key concepts are well addressed. This clas

Elizabeth Loftus later went on to apply this theory to cases

Costandi is a really good science writer who focuses on cog